



Assignment-3

Indefinite Integration

1. $\int (\sqrt{\cot x} e^{\sqrt{\sin x}} \sqrt{\cos x}) (1 + e^{\sqrt{\sin x}}) dx =$
 (A) $\frac{(1 + e^{\sqrt{\sin x}})^2}{2} + C$ (B) $2e^{\sqrt{\sin x}} + C$ (C) $\frac{e^{(1 + \sqrt{\sin x})}}{2} + C$ (D) $(1 + e^{\sqrt{\sin x}})^2 + C$
2. $\int \frac{\sec x(2 + \sec x)}{(1 + 2\sec x)^2} dx$
 (A) $\frac{\cos x}{(1 + 2\cot x)^2} + C$ (B) $\frac{\sin x}{(2 + \cos x)} + C$ (C) $\frac{-\sin x}{(2 + \cos x)^2} + C$ (D) $\frac{\csc x}{2 + \cos x} + C$
3. $P = 2016^{\left(\frac{\log_{10}\left(\frac{2015}{x} - 1\right)}{-\log_{10} 2016} \right)}$ then
 (A) $(x - 2015)\ln(2015 - x) - x \ln x + C$ (B) $(x - 2015) \ln(2015 - x) + x - x \ln x + C$
 (C) $(2015 - x) \ln(2015 - x) - x \ln x + C$ (D) $(2015 - x) \ln(2015 - x) + x \ln x + C$
4. $\int \frac{(1+x)[(1-x+x^2)(1+x+x^2)+x^2]}{(1+2x+3x^2+4x^3+3x^4+2x^5+x^6)} dx$
 $= f(x) + C$, C is integration constant, $f(0) = 0$, then $f(5) =$
 (A) 5 (B) $\ln 5$ (C) $\frac{1}{5} \ln 5$ (D) $\ln 6$
5. Let $f(x) = \tan x + 2 \tan 2x + 4 \tan 4x + 8 \cot 8x$ then primitive of $f(x)$ w.r.t. x is
 (A) $8 \ln |\sin 8x| + C$ (B) $\ln |\sin 8x| + C$ (C) $\ln |\sin x| + C$ (D) $\ln |\sin x| + C$
6. If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx =$
 $\log_e (f(x)) + g(x) + C$ and $f(x)$ is positive $f'(0) = 2$, $g(0) = 0$ then
 (A) $f(x) + g(x)$ is an even function
 (B) $f(x) - g(x)$ is an increasing function
 (C) $|f(0) + g(2)| = 1$
 (D) If a function $h(x)$ is such that $h(x) + e^x = f(x) + g(x)$ then maximum value of $h(x)$ is 1.
7. Let $I_1 = \int \frac{\sin^2 x + \sin x}{1 + \sin x + \cos x} dx$, $I_2 = \int \frac{\cos^2 x + \cos x}{1 + \sin x + \cos x} dx$ then which of the following is/are correct
 (A) $2I_1 + \sin x + \cos x - x = C$ (B) $I_1 + I_2 - x = C$
 (C) $I_1 - I_2 + \sin x + \cos x = C$ (D) $2I_2 + \sin x - x - \cos x = C$
8. If $\int \frac{1}{5\cos^2 x - 4\cos x \sin x - 2\sin 2x} dx = \frac{1}{2\sqrt{14}} \ln \left| \frac{k + \tan x + 1}{k - \tan x - 1} \right| + C$ then integral part of k^2 is
9. If $\int \cot^2 x \operatorname{cosec}^4 x dx = a \cot^3 x + b \cot^5 x + C$ then $\frac{1}{|a|} + \frac{1}{|b|}$ is equal to
10. If $f(x) = \int x^{\sin x} (\sin x + x \cos x \ln x + 1) dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$ then

(A) $f\left(\frac{\pi}{2}\right) < f\left(\frac{\pi}{4}\right)$ (B) $f\left(\frac{\pi}{6}\right) = \left(\frac{\pi}{6}\right)^{3/2}$ (C) $f\left(\frac{2}{3}\right) = \left(\frac{3}{2}\right)$ (D) $f(\pi) < f(2\pi)$

11. Let $f(x)$ be a differentiable function so that $f'(x)$ is continuous function and $\int \frac{(f(x) - f'(x))e^x}{(e^x + f(x))^2} dx = g(x) + C$ then which of the following is/are true

(A) $g(x) = \frac{1}{1+e^x}$ if $f(x) = e^{2x}$

(B) $g(\pi) = g(2\pi)$ if $f(x) = \sin x$

(C) $g'(x) = 0$ has atleast one solution in $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ if $f(x) = \cos x$

(D) $g(x)$ is a bounded function when $f(x) = e^x$

12. If $f(x) = \int \frac{2 + \sqrt{x}}{(x+1+\sqrt{x})^2} dx$ and $f(0) = 0$ and $g(x) = (x+q+\sqrt{x}) f(x)$ then

(A) Integral part of $g(4)$ is 8

(B) $f(9) = \frac{18}{13}$

(C) the graph of $g(x)$ intersect the line $y = 2x + 3$ at two points

(D) $\int_0^4 g(x) dx = 16$

Passage (13-14)

If $\int \frac{x^3 - 6x^2 + 11x - 6}{\sqrt{4x^2 + 4x + 3}} dx = (lx^2 + mn + n) \sqrt{x^2 + 4x + 3} + \lambda \int \frac{dx}{\sqrt{x^2 + 4x + 3}}$ then

13. Value of $n + \lambda =$

(A) $3l + 6m$

(B) $6m - 3l$

(C) $3l - 6m$

(D) $6l + 3m$

14. $\int_0^7 3(lx^2 + mx) - (\lambda + 17) dx$

(A) 0

(B) 49

(C)

(D)

Passage (15-16)

Let $g(x) = \int \sin(a+1)x (\sin x)^{a-1} dx = f(x) + C$, $g(0) = 0$, $a > 1$

15. $f\left(\frac{\pi}{2}\right)$ where $a = 50$ is

(A) 0

(B) -1

(C) $\frac{1}{50}$

(D) $-\frac{1}{50}$

16. Which of the following differs from other options when $a = 100$?

(A) $f\left(\frac{\pi}{5}\right)$

(B) $f\left(\frac{\pi}{10}\right)$

(C) $f\left(\frac{\pi}{4}\right)$

(D) $f\left(\frac{\pi}{15}\right)$

17. If $\int u \frac{d^2v}{dx^2} dx = u \frac{dv}{dx} - v \frac{du}{dx} + w$ then w is equal to (where u, v, w are differentiable function of x)

(A) $\int v \frac{d^2u}{dx^2} dx$

(B) $\int u \frac{d^2v}{dx^2} dx$

(C) $\int U \left(\frac{du}{dx}\right)^2 dx$

(D) $\int u \left(\frac{dv}{dx}\right)^2 dx$

18. If $F(x) = \int e^x \left(\int x \sin x^2 dx + \int (\sin x^2 + 2x^2 \cos x^2) dx \right) dx$ and $\lim_{x \rightarrow -\infty} F(x) = \frac{1}{2}$, then which of the following statement(s) must be true?

- (A) $F(0) = 0$ (B) $F'(0) = -\frac{1}{2}$ (C) $F(0) = 1$ (D) $F(0) = F'(0) = \frac{1}{2}$

19. $\int \sqrt{\frac{2-x}{2+x}} \cos^{-1} \left(\frac{\sqrt{2+x}}{2} \right) dx$

20. $\int \frac{(x^2 + 2) dx}{x \sqrt{(-x^4 + 5x^2 - 4)} \sin^{-1} \left(\frac{x^2 - 2}{x} \right)}$

21. $\int \frac{x^4 - 1}{\sqrt{2x^7 + x^4 + 6x^3}} dx$

22. $\int \frac{2x \sin^2 x - x^2 \sin 2x}{\sqrt{x^8 - x^6 \sin^2 x}} dx = \frac{\lambda \sqrt{x^2 - \sin^2 x}}{x} + C$, $x > 0$ (where c is integration constant) then λ can not be

23. if $\sin(ax) \sin(bx) = \cos(ax) \cos(bx)$ ($a, b \neq 0$ and $a \neq b$) then $\int \frac{\sin ax + \cos bx}{\cos ax + \sin bx} dx$ is

- (A) $\frac{1}{a} \ln |\sec ax| + C$ (B) $\frac{1}{b} \ln |\sin bx| + C$ (C) $\frac{1}{a} \ln |\operatorname{cosec} bx| + C$ (D) $\frac{1}{b} \ln |\cos ax| + C$

Passage (24-26)

Let $\int \frac{(x^4 + 2) dx}{(x^2 + 2)\sqrt{x^2 - 1}} = f(x) + C$ where $f(1) = 0$

24. $f(2)$ is equal to

25. Number of solutions of the equation $f'(x) = 1$ is equal to

26. Let $g(x) = \int x(2f(x) + f'(x)) dx$ then $g(-1)$ is equal to (where $g(1) = 0$)

27. Let $\int \frac{\sin^3 x dx}{1 + \sin^2 x} = \frac{1}{p} \ln \left| \frac{q + \cos x}{r - \cos x} \right| - \cos x + C$ then $(pq + pr)$ is equal to

28. If $\int \sin^{-1} \left(\frac{2x+1}{\sqrt{4x^2 + 4x + 10}} \right) dx = \left(\frac{2x+1}{2} \right) \tan^{-1} \left(\frac{2x+1}{3} \right) + \lambda \ln(4x^2 + 4x + 10) + C$ then

29. $\int \frac{(1 + \tan^2 x) dx}{1 - (\sin x \sqrt{\tan^2 x (\tan^2 x + 2) + 1})}$ where $x \in \left(\frac{9\pi}{2}, \frac{11\pi}{2} \right)$ is

30. If $\int (5 + 6x)(x^2 + x^3)^{1/3} x^5 dx = kx^e(1+x)^m + c$ then km is equal to

31. Let $\frac{d}{dx}(f(x)) = \frac{(\sin^{12} x - \cos^{12} x)}{(\sin^2 x - \cos^2 x)(1 - 3\sin^2 x \cos^2 x)} dx$ be such that $f(0) = 0$ then $f\left(\frac{\pi}{2}\right)$ equals $\frac{k\pi}{16}$. Find the value of k ($k \in \mathbb{N}$).

32. Let $\int \frac{(5x^4 + 1) dx}{3x^{4/3} \sqrt{x^4 - 1}} = \frac{\sqrt{f(x)}}{x^m} + C$ then $\frac{3}{m}$ is equal to

33. If $\int (x^6 + 7x^5 + 6x^4 + 5x^3 + 4x^2 + 3x + 1)e^x dx$ is equal to $\sum_{k=1}^{\alpha} \beta x^k - e^x + c$ (where c is constant of integration) then $(\alpha + \beta)$ is

34. If $\int \frac{(1 + x \cos x - \sin x) dx}{(x + \sin x + \cos x)^{1/2} (x - \sin x + \cos x)^{3/2}} = (x + l \sin x + m \cos x)^n \cdot (x + p \sin x + q \cos x)^r + c$

where $l, n > 0$ then $\left(\frac{l+m}{n}\right)(p-q)r$ is equal to

Passage (35-37)

For $x \in \left(0, \frac{\pi}{2}\right)$, let $f_n(x) = \int x \sin 2x (\sin^{2n-2} x - \cos^{2n-2} x) dx - \frac{1}{2^{n-1}}$, $n \in N$ and $f_n\left(\frac{\pi}{4}\right) = \frac{1}{2^{n-1}}$.

35. The derivative of $f_n(x)$ when $n = 2$ and $x = \frac{\pi}{3}$ is

- (A) 1 (B) $\frac{\sqrt{3}}{2}$ (C) 0 (D) $\frac{1}{2}$

36. If $g(x) = \lim_{n \rightarrow \infty} \sum_{n=1}^n f_n(x)$ then minimum value of $g(x)$ is

37. If $n = \lim_{t \rightarrow \infty} [\cot^{-1} t]$ then $\int \frac{f_3(x)}{\sin^2 x \cos^2 x} dx$ equals [.] is GIF

Passage (38-39)

Let $I_r(x) = \int \frac{x^r dx}{x^4 + 4}$ ($x > 0$)

38. If $I_1(\sqrt{2}) = \frac{9\pi}{16}$ then $\lim_{x \rightarrow 0} \frac{\cos(I_1(x))}{x^2}$ is equal to

39. $I_2(x) + 2I_0(x)$ is equal to

Passage (40-41)

Let $2I_2 + I_2 = \int \frac{e^x}{e^{2x} + e^{-2x}} dx$ and $I_1 + 2I_2 = \int \frac{e^{-x}}{e^{2x} + e^{-2x}} dx$.

40. I_1 is equal to

41. I_2 is equal to

Passage (42-43)

Let $p(x)$ be a polynomial function on R such that $p(x) + p(2x) = 5x^2 - 18$.

42. Number of solutions of the equation $p(x) = e^x$ is 43. $\int \frac{dx}{\sqrt{p(x)+25}}$ equals

Passage (44-45)

let $\int (2x^6 + 15x^4 + 2x^2 + 3) \cos 2x dx = f(x) \sin(2x) + g(x) \cos(2x) + k$ where $f(x)$ and $g(x)$ are polynomial function of x .

44. $\int \frac{g(x)}{f(x)} dx$ equals 45. $\int_{-1}^1 f(x) dx$ equals

Passage (46-47)

Let f, g, h be 3 functions such that $f'(x) > 0$ and $g(x) \geq 0 \quad \forall x \in R$ where $\int f(x) g(x) dx = \frac{x^4}{4} + C$ and

$$\int \frac{f(x)}{g(x)} dx = \int \frac{g(x)}{h(x)} dx = \ln|x| + C$$

46. Let $\int f(x) g(x) h(x) dx$ is equal to

47. $f(x), g(x), h(x)$ are in

- (A) A.P (B) G.P (C) H.P (D) NOT



Assignment-2

Indefinite Integration

- If $\int \frac{dx}{x\sqrt{1-x^3}} = a \log \left| \frac{\sqrt{1-x^3}-1}{\sqrt{1-x^3}+1} \right| + C$, then $a =$
(1) $1/3$ (2) $2/3$ (3) $-1/3$ (4) $-2/3$
- If $\phi(x) = \lim_{n \rightarrow \infty} \frac{x^n - x^{-n}}{x^n + x^{-n}}$, $0 < x < 1$, $n \in N$, then $\int \sin^{-1} x \phi(x) dx$ is equal to
(1) $x \sin^{-1} x + \sqrt{1-x^2} + C$ (2) $-(x \sin^{-1} x + \sqrt{1-x^2}) + C$
(3) $x \sin^{-1} x - \sqrt{1-x^2} + C$ (4) NOT
- The antiderivative of $\frac{\cos 5x + \cos 4x}{1 - 2 \cos 3x}$ is
(1) $\frac{\sin 2x}{2} + \cos x + C$ (2) $-\frac{\sin 2x}{2} + \sin x + C$
(3) $-\frac{\sin 2x}{2} - \sin x + C$ (4) $\frac{\sin 2x}{2} - \cos x + C$
- $\int \left(\frac{\ln x - 1}{(\ln x)^2 + 1} \right)^2 dx$ is equal to
(1) $\frac{x}{x^2 + 1} + C$ (2) $\frac{\ln x}{(\ln x)^2 + 1} + C$ (3) $\frac{x}{(\ln x)^2 + 1} + C$ (4) $e^x \left(\frac{x}{x^2 + 1} \right) + C$
- $\int_{-1}^{3/2} |x \sin \pi x| dx =$
(1) $\frac{3}{\pi} - \frac{1}{\pi^2}$ (2) $\frac{3}{\pi} + \frac{1}{\pi^2}$ (3) $\frac{2}{\pi} + \frac{1}{\pi^2}$ (4) $\frac{3}{\pi} + \frac{1}{\pi}$
- Let $f(x) = \max(x + |x|, x - [x])$ where $[.]$ denotes GIF. Then $\int_{-2}^2 f(x) dx$ is equal to
(1) 1 (2) 0 (3) 4 (4) 5
- If $f(x) = \int_0^x \frac{dt}{(f(t))^2}$ and $\int_0^2 \frac{dt}{(f(t))^2} = \sqrt[3]{6}$ then $f(9) =$
(1) 0 (2) 1 (3) 2 (4) 3
- The greatest value of $f(x) = \int_1^x |t| dt$ on the interval $\left[-\frac{1}{2}, \frac{1}{2}\right]$ is
(1) $\frac{3}{8}$ (2) $-\frac{3}{8}$ (3) $-\frac{1}{2}$ (4) $\frac{1}{2}$
- If $\int \frac{4e^x - 6e^{-x}}{9e^x - 4e^{-x}} dx = Ax + B \ln(9e^{2x} - 4) + C$ then $A + B =$
(1) $-\frac{9}{35}$ (2) $-\frac{19}{36}$ (3) $\frac{19}{36}$ (4) $\frac{9}{35}$

10. $\int \sin 51x (\sin x)^{49} dx$ equals
 (1) (2) (3) (4)
11. Let $f(x)$ be a cubic polynomial with leading coefficient unity such that $f(0) = 1$ are all the roots of $f'(x) = 0$ are also roots of $f(x) = 0$. If $\int f(x) dx = g(x) + C$ where $g(0) = \frac{1}{4}$ and c is constant of integration, and then $g(3) - g(1)$ is equal to
 (1) 27 (2) 48 (3) 60 (4) 81
12. $\int \frac{e^x(x-1)(x-\ln x)}{x^2} dx$ equals to
 (A) $e^x \left(\frac{x - \ln x}{x} \right) + C$ (B) $e^x \left(\frac{x - \ln x + 1}{x} \right) + C$
 (C) $e^x \left(\frac{x - \ln x}{x^2} \right) + C$ (D) $e^x \left(\frac{x - \ln x - 1}{x} \right) + C$
13. If $x \in (0, 1)$ then
 $\int \sqrt{x} \tan \left(2 \tan^{-1} \left(\frac{\sqrt{1+\sqrt{x}} + 1 - \sqrt{1+\sqrt{x}} - 1}{\sqrt{1+\sqrt{x}} + 1 + \sqrt{1+\sqrt{x}} - 1} \right) \right) dx$ is equal to
 (A) $\frac{2}{3}x^{3/4} + C$ (B) $\frac{4}{3}x^{3/4} + C$ (C) $\frac{4}{5}x^{5/4} + C$ (D) $\frac{2}{5}x^{5/4} + C$
14. If $\int (x^{24} + x^{16} + x^8) (2x^{16} - 3x^8 + 6)^{1/8} dx = c + \frac{1}{\alpha} (2x^{24} + 3x^{16} + 6x^8)^{\beta/\gamma}$ (β, γ are coprime natural numbers) then the value of $\alpha + \beta + \gamma$ is
 (A) 51 (B) 61 (C) 71 (D) 81
15. $\int \operatorname{cosec}^6 x dx$ is equal to
 (A) $-\cot x - \frac{\cot^5 x}{5} - \frac{2\cot^3 x}{3} + k$ (B) $-\frac{\cot x}{3} + \frac{2\cot^5 x}{5} + 2\cot^{-3} x + k$
 (C) $\frac{\tan^3 x}{3} - \frac{\tan x}{5} + 2\tan^3 x + k$ (D) NOT
16. $\int \frac{(\sqrt{x^2+1}) \{ \ln(x^2+1) - 2 \ln x \}}{x^4} dx$ is equal to
 (A) $\frac{(x^2+1)\sqrt{x^2+1}}{x^3} \left[2 - 3 \ln \left(\frac{x^2+1}{x^2} \right) \right] + c$ (B) $\frac{1}{9} \frac{(x^2+1)\sqrt{x^2+1}}{x^3} \left[2 + 3 \ln \left(\frac{x^2+1}{x^2} \right) \right] + c$
 (C) $\frac{(x^2+1)\sqrt{x^2+1}}{x^3} \left[2 + 3 \ln \left(\frac{x^2+1}{x^2} \right) \right] + c$ (D) $\frac{1}{9} \frac{(x^2+1)\sqrt{x^2+1}}{x^3} \left[2 - 3 \ln \left(\frac{x^2+1}{x^2} \right) \right] + c$
17. $\int \frac{x^3 - 3x + 7}{x^2 + 4} dx$ is equal to
 (A) $\frac{x^2}{2} + \frac{7}{2} \ln(x^2 + 4) + C$ (B) $\frac{x^2}{2} + \frac{7}{2} + \tan^{-1} \frac{x}{2} - \frac{7}{2} \ln(x^2 + 4) + C$
 (C) $-\frac{x^2}{2} + \frac{7}{2} \tan^{-1} \frac{x}{2} + \frac{7}{2} \ln(x^2 + 4) + C$ (D) $\frac{x}{2} + \frac{7}{2} \tan^{-1} \frac{x}{2} + C$

18. $I_{m,n} = \int_0^1 x^m (\ln x)^n dx =$
- (A) $-\frac{m+1}{n+1} I_{m,n+1}$ (B) $-\frac{m}{n+1} I_{m,n-1}$ (C) $-\frac{n}{m+1} I_{m,n-1}$ (D) $\frac{m}{n+1} I_{m,n-1}$
19. $\int \frac{\sin 2x - \sin 2\alpha}{\sin x - \sin \alpha - \cos x + \cos \alpha} dx =$
- (A) $\sqrt{2} \sin\left(x + \frac{\pi}{4}\right) + \sqrt{2} \sin\left(\frac{\pi}{4} - \alpha\right)x + C$ (B) $\sqrt{2} \cos\left(\frac{\pi}{4} - x\right) = \sqrt{2} \sin\left(\alpha - \frac{\pi}{4}\right) + C$
- (C) $\sqrt{2} \cos\left(x + \frac{\pi}{4}\right) + \sqrt{2} \cos\left(\frac{\pi}{4} - \alpha\right)x + C$ (D) $\sqrt{2} \sin\left(-x + \frac{\pi}{4}\right) + \sqrt{2} \cos\left(\frac{\pi}{4} + \alpha\right)x + C$
20. If $I = \int \frac{e^{2x} + e^{-2x} + 1}{e^{-2x} + e^{-x} + 1} dx = \frac{e^{9x}a}{a} - \frac{e^{bx}}{b} + \frac{x}{c} + C$ where $a, b, c \in \mathbb{N}$, then the value of $b^{a-c} + c^{c-b} + a^{a-b} =$
21. $I_1 = \int \tan x + \tan(ax+b) dx$ and $I_2 = \int \cot x \cot(ax+b) dx$, where a, b are non-zero real numbers.

Column I

- (A) Value of I_1 for $a = 1$ is
- (B) Value of I_2 for $a = 1$ is
- (C) Value of I_1 for $a = -1$ is
- (D) Value of I_2 for $a = -1$ is

Column II

- (P) $x - \cot b \ln\left(\frac{\cos(x-b)}{\cos x}\right) + C$
- (Q) $\cot b \ln\left(\frac{\sin x}{\sin(b+x)}\right)$
- (R) $\cot b \ln\left(\frac{\cos x}{\cos(x+b)}\right) - x + C$
- (S) $x + \cot b \ln\left(\frac{\sin x}{\sin(b-x)}\right) + C$

22. $\int \frac{\sin x + \sqrt[3]{\sin^2 x} + \sqrt[6]{\sin x}}{(1 + \sqrt[3]{\sin x}) \tan x} dx$
- (A) $\frac{3}{2}(\sin x)^{2/3} + \tan^{-1}(\sin^6 x) + C$ (B) $\frac{3}{2}(\sin x)^{2/3} + 6 \tan^{-1} \sin^6 x + C$
- (C) $\frac{3}{2}(\sin x)^{2/3} + 6 \tan^{-1}(\sin^{1/6} x) + C$ (D) $\frac{\sin^2 x + 1}{(\sin x)^{2/3}} + C$
23. If $f(x)$ is a polynomial function such that $f(x) + f'(x) + f''(x) + f'''(x) = x^3$ and $g(x) = \int \frac{f(x)}{x^3} dx$ and $g(1) = 1$ then $g(e) =$
- (A) $e + 3$ (B) $e - 3$ (C) $1 + e$ (D) $1 - e$
24. If $\int (\sin 3\theta + \sin \theta) e^{\sin \theta} \cos \theta d\theta = (A \sin^3 \theta + B \cos^2 \theta + C \sin \theta + D \cos \theta + E) e^{\sin \theta} + F$ then
- (A) $B = 12$ (B) $D = 0$ (C) $B = -12$ (D) $A = 4$
25. Consider the function $h(x) = \frac{g^2(x)}{2} + 3x^3 - 5$ where $g(x)$ is a continuous and differentiable function. If is given that $h(x)$ is a monotonically increasing function and $g(0) = 4$, then
- (A) $g^2(1) > 1$ (B) $h(5) > 3$ (C) $h\left(\frac{5}{2}\right) < 2$ (D) $g^2(-1) < 22$

26. If $\int \frac{\cos x}{x} (x^2 \ln x + 1) dx = f(x) + C$
 $f(1) = \cos 1$ and $\int f(x) \sec^2 x dx = g(x) + C$ $g(1) = 0$ and $L = \lim_{x \rightarrow 0^+} g(x)$ then
- (A) $f(x) = \cos x(1 + \ln x) + x \sin x \ln x$ (B) $g(x) = \frac{x \ln x}{\cos x}$
 (C) $L = 0$ (D) $\lim_{x \rightarrow \infty} \frac{g(x)}{x^2} = 0$
27. If $\int \frac{(1+x)((1-x+x^2)(1+x+x^2)+x^2)}{(1+2x+3x^2+4x^3+3x^4+2x^5+x^6)} dx = g(x) + C$ and $g(1) = \ln 2$ then which of the following is/are true?
- (A) $g(x)$ is an increasing function $\forall x > -1$
 (B) $[g(99) - g(3)] = 3$ where $[.]$ is GIF
 (C) $g(x)$ is a periodic function
 (D) $g(6) - g(7) > 0$
28. If $\int \frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = C - \frac{1}{f(x)}$ where $f(0) = 1$ and C is integration constant then the value of $f(1) =$ _____.
29. $\int \frac{(x^{3x} + x^x)(\log x + 1)}{x^{4x} + 1} dx = ('C' \text{ is integration constant})$
- (A) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^{2x} - 1}{\sqrt{2}x^x} \right) + C$ (B) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^x + 1}{\sqrt{2}x^x} \right) + C$
 (C) $\sqrt{2} \tan^{-1} \left(\frac{x^{2x} + 1}{x^x} \right) + C$ (D) $\sqrt{2} \tan^{-1}(x^x + 1) + C$
30. If A is a square matrix and $e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$ and $e^A = \frac{1}{2} \begin{bmatrix} f(x) & g(x) \\ g(x) & f(x) \end{bmatrix}$ and $A = \begin{bmatrix} x & x \\ x & x \end{bmatrix}$, $0 < x < 1$, I is a 2×2 identity matrix then which of the following statement is/are true? (C is constant)?
- (A) $\int \frac{g(x)}{f(x)} dx = \log(e^x + e^{-x}) + C$
 (B) $\int \frac{f(x) - g(x)}{f(x) + g(x)} dx > 2e^{-2x} + C$
 (C) $\int (g(x) + 1) \sin x dx = \frac{e^{2x}}{5} (2 \sin x - \cos x) + C$
 (D) $f(x) + g(x)$ is an increasing function $\forall x \in R$
31. $\int \frac{((\tan x - 1)^3 (\tan x + 2)^5)^{-1/4}}{\cos^2 x} dx = \frac{a}{b} \left(\frac{\tan x - 1}{\tan x + 2} \right)^{c/4} + k$ where k is integration constant a, b, c are relatively prime positive integers then
- (A) $a + b - c = 6$ (B) $ab - c = 11$ (C) $a^c + b^c = 17$ (D) $a - b - c = 0$

Passage (32-33)

Let f be a differentiable function f satisfies the relation

$f(xy) = f(x) + f(y) + xy - x - y \forall x, y > 0, f'(1) = 4$ then answer the following

32. $\int \frac{f(x)}{x} dx =$

(A) $3(\ln x)^2 + x + c$

(B) $3\ln x + x + c$

(C) $\frac{3}{2}\ln x + x + c$

(D) $\frac{3}{2}(\ln x)^2 + x + c$

33. $\int_2^3 (e^{f(x)-x} + x) dx$

(A) 0

(B) $\frac{75}{4}$

(C) $\frac{1}{4}$

(D) $\frac{75}{2}$

Answer

1. (3)

2. (2)

3. (3)

4. (3)

5. (2)

6. (4)

7. (4)

8. (2)

9. (2)

10. (2)

11. (3)

12. (D)

13. (C)

14. (C)

15. (A)

16. (D)

17. (B)

18. (AC)

19. (AB)

20. (4)

21. A-R, B-Q, C-P, D-S

22. (A)

23. (B)

24. BCD

25. ABD

26. ABCD

27. AB

28. 5

29. A

30. ACD

31. ABD

32. D

33. B